MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026
FIFTH SEMESTER
Discipline Specific Core Course (DSC)- MG5DSCZGY302, MG5DSCZIM301-
FUNDAMENTALS OF GENETICS
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A
MCQ

Answer **all** questions. Each question carries **1** mark

1. What phenotypic ratio is obtained when the F1 hybrid Tt is crossed with the recessive parent tt?
   (d) 1 tall: 1 dwarf [A]/[CO1]
2. A gene in Drosophila affects wing size, eye colour, fertility and lifespan. Which genetic phenomenon does this situation demonstrate? [A]/[CO1]
   (c) Pleiotropism
3. In roan cattle, both red and white hairs are expressed. This illustrates: [U]/[CO1]
   (b) Co-dominance
4. Recombination frequency is expressed as**:** [K]/[CO2]
   (c) Percentage recombinant offspring
5. A Drosophila has an X/A ratio of 1.0. Which sex type is expected? [A]/[CO2]
   (b) Normal female
6. A frameshift mutation changes the: [K]/[CO3]
   (c) Reading frame
7. Which symbol represents a male in a human pedigree? [K]/[CO4]
   (b) Square
8. Which type of chromatin isa generally less condensed? [U]/[CO4]
   (c) Euchromatin
9. Which of the following is commonly associated with Klinefelter's syndrome? [U]/[CO5]
   (a) Gynaecomastia
10. Cleft lip primarily involves a defect in the: [K]/[CO5]
    (b) Upper lip

PART B
Short Answer Questions

Answer any **10** questions. Each question carries **3** marks

11. Distinguish between complementary and supplementary gene interaction. [U]/[CO1]

Complementary gene interaction	Supplementary gene interaction
Two or more dominant genes interact with one another to produce a single trait.	One dominant gene produces its effect in the presence or absence of the other dominant gene but the second one produces its effect only in the presence of first and a new trait is expressed when both are expressed together.

Complementary gene interaction	Supplementary gene interaction
If either gene is homozygous recessive, the trait is absent.	If the primary gene is recessive, it completely masks the effect of the other gene.
Dihybrid ratio- 9:7	Dihybrid ratio- 9:3:4
Eg. Flower colour in sweet peas	Eg. Inheritance of coat colour in mice

(Complementary and supplementary gene interactions- 1.5 marks each)

12. Define homozygous and heterozygous conditions with examples. [K]/[CO1]

Homozygous condition- An individual is homozygous for a specific gene when they inherit two identical alleles for that trait from both parents. Types- 1. Homozygous dominant: Has two dominant alleles. Eg. TT for tall plants; 2. Homozygous recessive: contains two recessive alleles.
Eg. tt for dwarf plants.
Heterozygous condition- An individual is heterozygous for a specific gene when they inherit two different alleles for that trait-one dominant and other recessive and the dominant trait is expressed. Eg. Tt (Tall plants). **(Homozygous and Heterozygous conditions- 1.5 marks each)**

13. What is meant by sex-influenced inheritance? How does it differ from sex-linked inheritance? [U]/[CO1]

Sex influenced genes are genes influenced by the sex of the bearer. Found in both sexes. Controlled by autosomal genes, sex hormones act as modifiers. Eg. Male pattern baldness.
Sex linked inheritance refers to the transmission of traits encoded by genes located on the sex chromosomes-X-linked and Y-linked traits. Eg. Red-green colour blindness, Hemophilia A, Duchenne muscular dystrophy. **(Sex-influenced and Sex-linked inheritance- 1.5 marks each)**

14. Demonstrate the possible offspring genotypes produced in the cross. RrYy x rryy. [U]/[CO1]

RrYy x rryy

P **RrYy** **rryy**

G (RY) (Ry) (rY) (ry) (ry)

F1

	RY	**Ry**	**rY**	**ry**
ry	RrYy	Rryy	rrYy	rryy

Possible offsprings: RrYy, Rryy, rrYy and rryy. **(Cross with result- full marks)**

15. Describe the sexual dimorphism in *Bonellia viridis*. [K]/[CO2]

In *Bonellia viridis*, sex is determined by environmental factors. It exhibits sexual dimorphism. Female is very large with proboscis and lives in a burrow at sea bottom. Male is very small and lives as a parasite in the uterus of adult female. Larvae of Bonellia are alike and they have the potentialities to develop into any sex. Larva settled on proboscis of females develop into male as proboscis secretes a hormone like substance that influences the larvae towards maleness.

(Male and female sexual dimorphisms- 1.5 marks each)

16. Explain why a freemartin is usually sterile. [U]/[CO2]

When twin calves of opposite sex are born, one is normal male and the other is usually a sterile female with male like traits and is called 'Free martin'. During early development, the fusion of blood vessels allows the blood of one embryo to enter the blood stream of other. Male hormone influence the development of female embryo into male direction. Because male sexual differentiation occurs earlier than female differentiation, male hormones- primarily Anti-Mullerian Hormone (AMH) and testosterone produced by the developing testes of the male twin cross into the female twin's bloodstream and suppress her reproductive development. This exposure leads to the degeneration of her Mullerian ducts and arrests ovarian development, leaving her with rudimentary or absent internal reproductive organs, non-functional ovaries and XX/XY leukocyte chimerism, ultimately rendering her permanently sterile. **(Brief account- Liberal valuation)**

17. Identify and explain the significance of a darkly stained chromatin body in the nucleus of a female somatic cell. [A]/[CO2]

The darkly stained chromatin body found in the nucleus of a female somatic cell is known as a Barr body (or sex chromatin). which represents a heavily condensed and transcriptionally inactive X chromosome. Its biological significance lies in dosage compensation, a mechanism that equalizes X-linked gene expression between males (XY) and females (XX) by randomly silencing one of the two X chromosomes during early embryonic development. This process ensures that females do not produce double the amount of X-linked gene products compared to males, maintaining proper cellular homeostasis. **(Identification- 1 mark, significance- 2 marks)**

18. Explain why an A G substitution is classified as a transition, whereas an A T substitution is classified as a transversion. [U]/[CO3]

An A to G substitution is classified as a transition because it involves the replacement of one purine (adenine, possessing a double-ring structure) with another purine (guanine), thereby preserving the same chemical ring-structure category. In contrast, an A to T substitution is classified as a transversion because it involves the replacement of a purine (adenine) with a pyrimidine (thymine, a single-ring structure), shifting between two entirely different chemical categories of nitrogenous bases. **(Transition- 1.5 marks, Transversion- 1.5 marks)**

19. Describe how a mutation occurring in a somatic cell can become restricted to a particular tissue or cell lineage. [U]/[CO3]

When a mutation occurs in a somatic cell post-zygotically, it is passed down exclusively to all mitotic progeny derived from that original cell through subsequent cell divisions. As embryonic development or adult tissue renewal progresses, progenitor and stem cells undergo clonal expansion to populate specific organs and structures. Consequently, if the mutation happens within a progenitor cell destined for a particular lineage, all descendant cells within that specific tissue will inherit the mutation, while the rest of the body's tissues remain unaffected because somatic mutations are confined locally and are not transmitted to other independent cell lineages or to offspring.

(Liberal valuation)

20. Define mutagens and list their major types with one suitable example each. [K]/[CO3]

Environmental or external agents that induce mutations are called mutagens. It is divided into two major groups: **1. Physical Mutagens**-(a) Ionising radiations- radiations with short wavelengths and strong energy like X-rays, protons, neutrons, alpha, beta and gamma rays from radioactive sources. (b) Non-ionizing radiations: UV rays results in pyrimidine hydrates and pyrimidine dimers (Thymine dimers). **2. Chemical Mutagens**- (a) Chemicals mutagenic to replicating and non replicating DNA- Eg. Alkylating agents such as nitrogen and sulphur mustards, methyl and ethyl methanesulfonates- add ethyl and methyl groups to bases and alter the base pairing potentials (7-ethyl guanine base pairs with thymine), Hydroxylating agents, nitrous acid (deamination of adenine to hypoxanthine and cytosine to uracil). (b) Chemical mutagenic only to replicating DNA. Eg. Acridine dyes and Base analogs (5-bromo uracil: thymine analog and 2-amino purine : adenine analog).

(Definition- 1 mark, Physical and chemical mutagens with example-1 mark each)

21. List the major levels of chromosome organization. [K]/[CO4]

Each chromosome consists of a single, enormously long linear DNA molecule associated with histone proteins that fold and pack the fine DNA thread into a compact structure called chromatin. Nucleosomes is the basic structural and functional unit of chromatin. Nucleosome core consists of a complex of eight histone proteins-two molecules each of histone H2A, H2B, H3 and H4 and double stranded DNA that is 146 nucleotides long. The formation of nucleosomes converts a DNA molecule into a chromatin thread about one-third of its initial length and provides first level of DNA packaging. H1 pulls nucleosomes together into a 30-nm fiber (40 fold, second level). 30 nm fiber folds into a series of radial loops and organizes chromatin into distinct functional sections within the interphase (third level). Chromatin is folded more than 100 fold during mitosis or meiosis, the looped domains undergo additional coiling and compression assisted by condensing and cohesion proteins (fourth level). **(3/4 marks for each levels- full marks)**

22. Explain the significance of sequencing the human genome. [U]/[CO5]

It helps to map and understand all the genes of human being. Its significance lies in enabling scientists to pinpoint the genetic basis of thousands of inherited diseases, trace human evolutionary history, and accelerate the development of personalized medicine, targeted therapies and advanced diagnostics that continue to transform modern healthcare. **(3 points- full marks)**

PART C
Short Essay Questions

Answer any **6** questions. Each question carries **5** marks.

23. A woman with blood group A gives birth to a child with blood group AB. Two men are considered possible fathers: one has blood group B and the other has blood group O. Which man could genetically be the father? Justify your answer. [A]/[CO1]

I case. Mother A group, Father B group.

I^A or I^o x I^B or I^o

	I^B	I^o
I^A	$I^A I^B$ AB group	$I^A I^O$ A group
I^o	$I^B I^O$ B group	$I^O I^O$ O group

II case. Mother A group, Father O group

I^A or I^o x I^O

	I^o
I^A	$I^A I^o$ A group
I^o	$I^o I^O$ O group

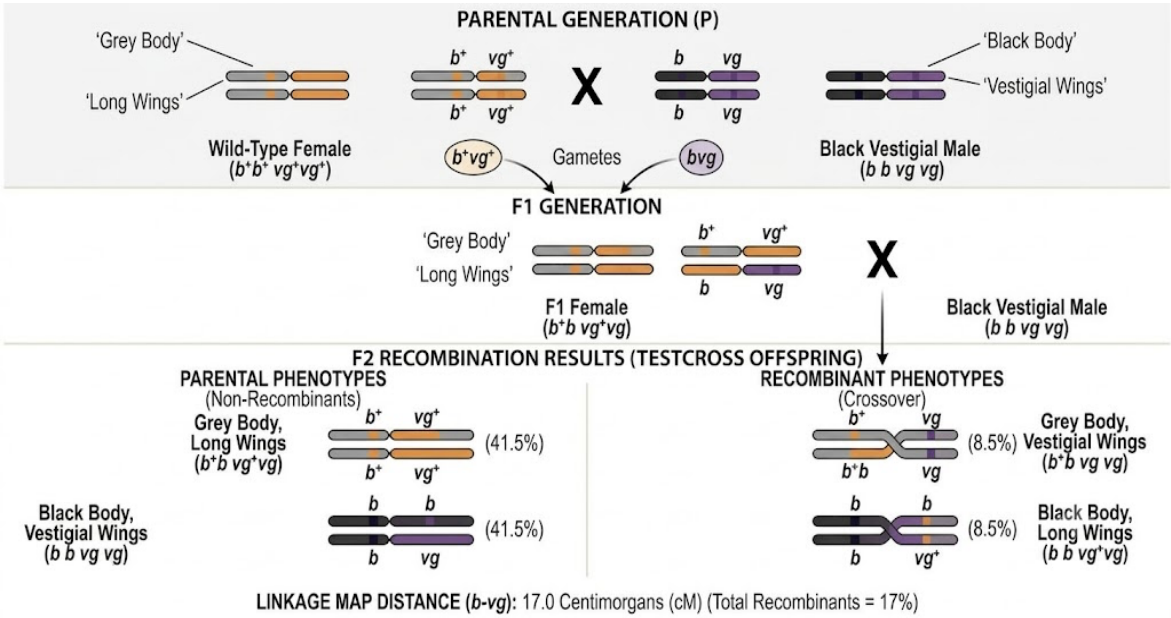
The man with blood group **B** is the father of the child because a cross between A group of mother and B group of man can only result in a child with AB group.

(Illustration- 2 marks, Result with reason-3 marks)

24. List the seven pairs of contrasting characters studied by Mendel in pea plants, indicating the dominant and recessive trait in each pair. [K]/[CO1]

No.	Character	Dominant	Recessive
1	Seed shape	Round	Wrinkled
2	Seed colour	Yellow	Green
3	Flower colour	Violet	White
4	Pod shape	Full	Constricted
5	Pod colour	Green	Yellow
6	Flower position	Axial	Terminal
7	Stem height	Tall	Dwarf

25. Describe Morgan's experiments in Drosophila involving grey-bodied, long-winged and black-bodied, vestigial-winged flies, with a neat labelled illustration, and explain his findings. [U]/[CO2]



Morgan's findings revealed that the genes for body color and wing size are physically located on the same chromosome and exhibit genetic linkage, meaning they tend to be inherited together as parental combinations rather than assorting independently as predicted by Mendelian principles. However, the appearance of a small percentage of recombinant offspring (17%) demonstrated that linkage is incomplete, as homologous chromosomes undergo crossing over during meiosis to exchange genetic material and produce novel allele combinations. Furthermore, Morgan and his colleagues established that the frequency of these recombinants correlates directly with the physical distance between the genes. He showed that coupling and repulsion are two phases of linkage corresponding to linkage and crossing over. **(Explanation- 3 marks, Diagram- 2 marks)**

26. Describe chromosomal sex determination in humans, indicating the sex-chromosome constitution and gamete types of males and females. [U]/[CO2]

In man, the chromosomal mechanism of sex determination is of **XX-XY** type. The total compliment of human chromosomes includes 44 autosomes and 2 sex chromosomes. Female is homogametic, XX, since all gametes are same and produces only one type of egg (22+X) while male is heterogametic, producing two different gametes, (22+X) and (22+Y). The sex of the offspring is determined during fertilisation by the kind of sperm that fuses with the egg. The X bearing sperm produces females (XX) and the Y bearing ones produces males (XY). The testis determining factor (TDF) gene in Y chromosome exhibit a dominant effect for the expression of maleness. A single Y chromosome is sufficient to produce testis and male characteristics.

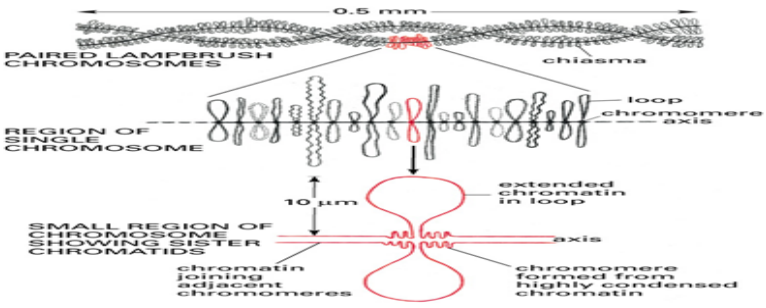
(Chromosomal Sex type in man- 3 marks, Male-1 mark, Female- 1 mark)

27. List the major numerical chromosomal aberrations and state their basic characteristics. [K]/[CO3]

Numerical aberrations: **Aneuploidy**: Condition in which one or few extra chromosomes are added or one or few chromosomes are missing from the basic chromosome sets. Usually arise through non-disjunction of chromosomes in meiosis. Two types: Hypoploidy- Aneuploidy involving loss of chromosomes. Eg. Monosomy-(2n-1), Nullisomy (2n-2). Hyperploidy- Involves addition of chromosomes. Eg. trisomy (2n+1)-Down syndrome (trisomy 21)- autosomal trisomy. Aneuploidy of autosomes and sex chromosomes have been reported in man. Examples for Aneuploidy and autosomal abnormalities. Down syndrome- most common autosomal abnormalities in man, due to autosomal trisomy of chromosome number 21, the affected individuals have 47 chromosomes (2n+1) which is due to non-disjunction that occurs in the meiotic division of either parent. Edward syndrome- trisomy 18. Patau syndrome- Trisomy13. **Euploidy,** which involves alterations in the number of complete sets of chromosomes. Within euploidy, variations include monoploidy (possessing a single set of chromosomes, n), usually occur in higher plants and polyploidy (possessing three or more complete sets, such as triploidy or tetraploidy), both of which alter entire genomic sets. Polyploids can be autopolyploids- when same genome is duplicated or allopolyploids- multiplication of chromosome sets that are initially derived from two different species. **(Types- 2.5 marks, Characteristics-2.5 marks)**

28. With the help of a labelled diagram, describe the structure of a lamp brush chromosome. [U]/[CO4]

Lampbrush chromosomes are exceptionally large meiotic chromosomes found predominantly during the diplotene stage of prophase I in vertebrate oocytes, characterized structurally by a central backbone of tightly condensed sister chromatids punctuated by darkly staining granules called chromomeres. Extending perpendicularly and symmetrically from these chromomeres are pairs of lateral chromatin loops that are actively undergoing transcription, giving the chromosome a brush-like appearance. There are two types of loops- Complex and plain loops. These chromosomes are highly active in RNA synthesis. The loops consist of transcriptionally active DNA which can synthesize large amount of mRNA.



(Description- 3 marks, diagram- 2 marks)

29. Compare phenylketonuria, alkaptonuria and albinism with respect to the defective enzyme or biochemical process, metabolic consequence and major clinical manifestations. [U]/[CO5]

Features	**Phenylketonuria (PKU)**	**Alkaptonuria (AKU)**	**Albinism**
Defective Enzyme / Gene	Phenylalanine hydroxylase (PAH)	Homogentisate 1,2-dioxygenase (HGD) or Homogentisic acid oxidase	Tyrosinase (TYR)
Biochemical Process Blocked	Conversion of phenylalanine to tyrosine	Breakdown of homogentisic acid to maleylacetoacetic acid	Conversion of tyrosine to DOPA/dopaquinone (melanin synthesis pathway)
Metabolic Consequence	Accumulation of phenylalanine and its ketoacids (e.g., phenylpyruvate) in blood/tissue; deficiency of tyrosine.	Accumulation and urinary excretion of homogentisic acid (HGA), which oxidizes upon standing.	Impaired or complete absence of melanin production in melanocytes.
Major Clinical Manifestations	Severe intellectual disability,microcephaly, Musty" or "mousy" odor, Light pigmentation (hypopigmentation), seizures	Urine darkens/turns black upon exposure to air, Ochronosis (bluish-black pigmentation of connective tissue/ cartilage), Early-onset severe arthritis	Absence/reduction of pigment in skin, hair, and eyes, Photophobia and nystagmus, Reduced visual acuity, High risk of skin cancers (e.g., melanoma)

(Defective enzyme- 1mark, Biochemical process- 1 mark, Metabolic consequence- 1 mark, Clinical manifestation- 2 marks)

30. Briefly explain the genetic and physiological basis of cystic fibrosis. [U]/[CO5]

Cystic fibrosis (CF) is an autosomal recessive genetic disorder. It is characterized by accumulation of mucus in the lungs, pancreas and liver, resulting in chronic infections and malfunction of these vital organs. CF gene was mapped to long arm of chromosome 7. Cystic Fibrosis Transmembrane conductance Regulator (CFTR) is a CF gene product and forms ion channels through the membrane of cells that line respiratory tract, pancreas, sweat glands, intestine and other organs and regulates the flow of water and salts across the membrane. Although CFTR forms chlorine channels, it also regulates the activity of several other transport system such as potassium and sodium channels. Since mutant CFTR does not function properly in CF patients, salt accumulates in epithelial cells and mucus builds upon surfaces of these cells. The presence of mucus on the lining of respiratory tract leads to chronic, progressive infections by *Pseudomonas aeruginosa*, *Staphylococcus aureus* and related bacteria. These infections results in respiratory failure and death. CF genes are also pleiotropic and therefore malfunctioning of pancreas, liver, bones and intestinal tract are common in individuals with CF. CFTR also plays a role in regulating lipid metabolism and transport. 70% of cases of CF are due to ΔF 508 trinucleotide deletion. **(Genetic basis- 2.5 marks, Physiological manifestations- 2.5 marks).**